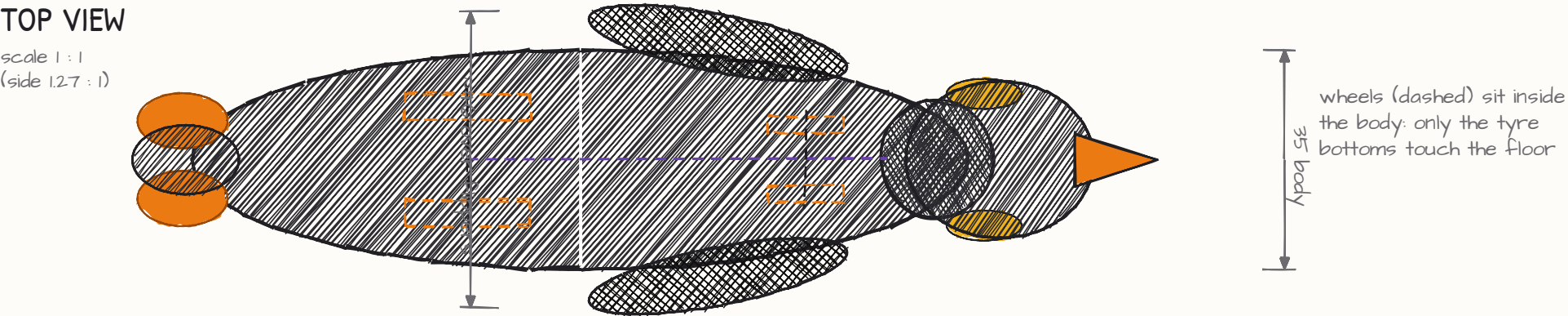
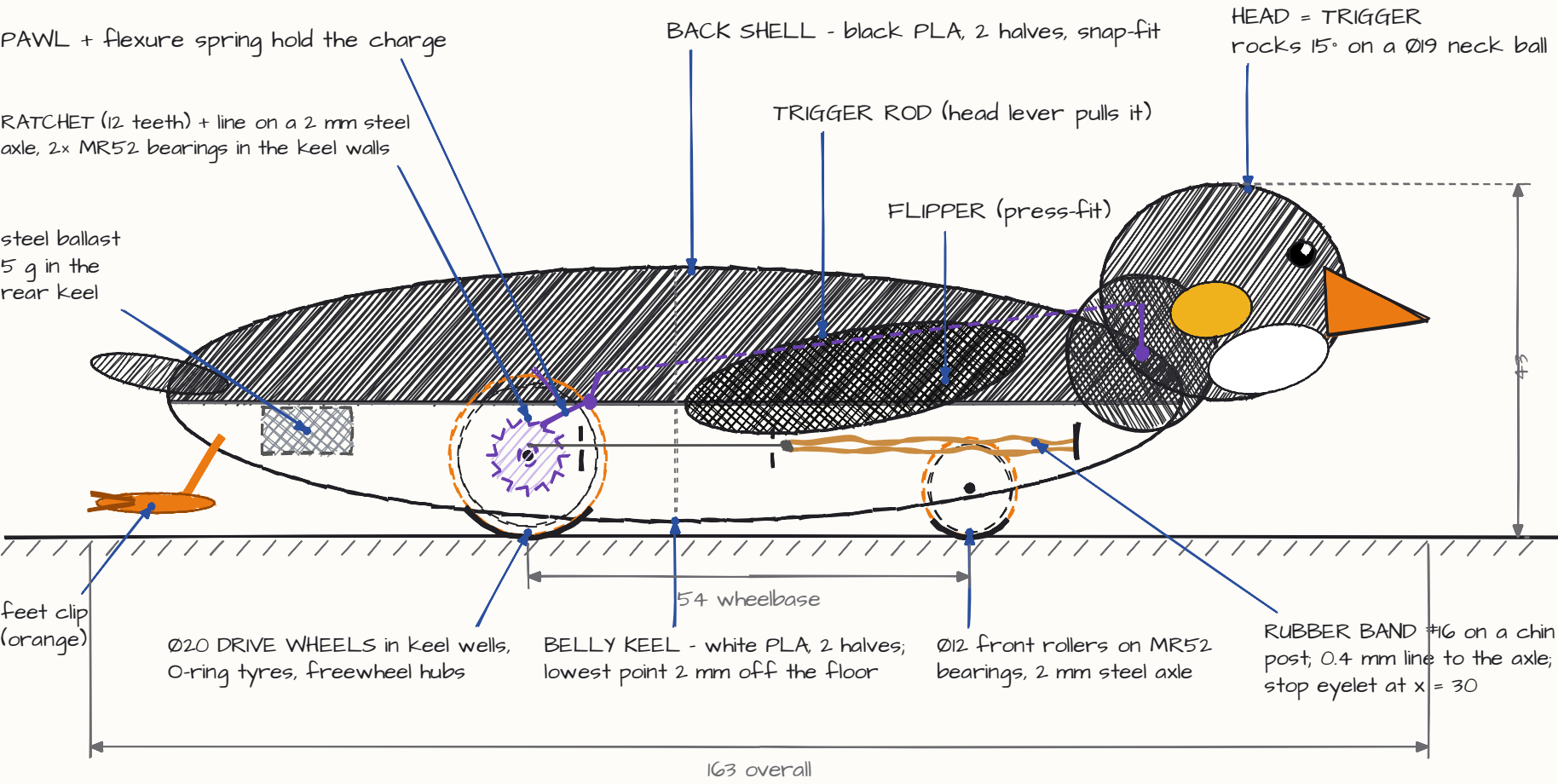
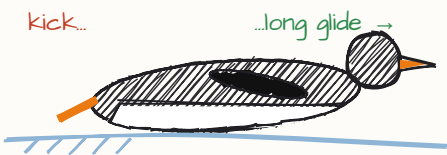


SIDE VIEW (scale 1:2.7 : 1 - dashed = hidden parts inside the body - purple = mechanism)



BIOMIMICRY



Tobogganing penguins (Emperor, Adélie) give a few kicks, then glide far on their bellies. Copied: a 22 cm power stroke (the band), then a long freewheel belly-glide on a keel; low, long body.

HOW IT PLAYS

- 1 Press on its back and pull back ~22 cm (click-click...)
- 2 Let go - the ratchet holds it; it waits on the floor.
- 3 Tap the head - it toboggans ~ 4 m across the floor.

vs Concept 2: lying body, pull-back band head-nod trigger.

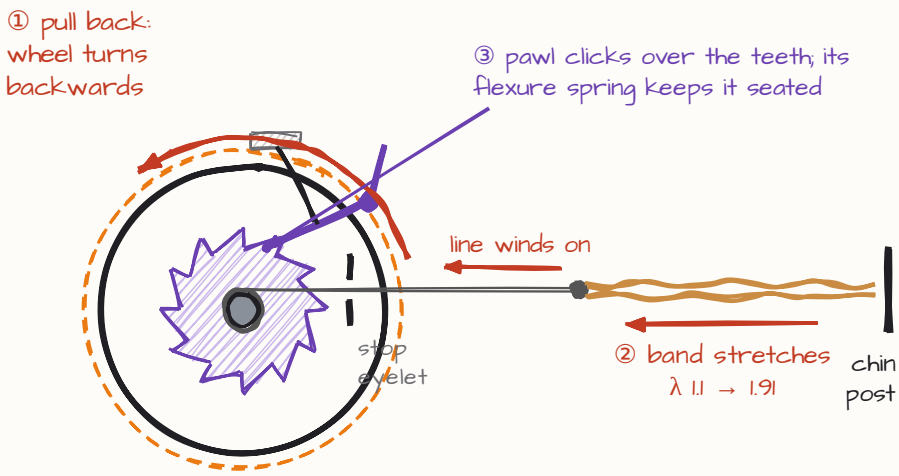
NUMBERS

- Mass ≈ 39 g, 78 % printed PLA
- CG 17 mm ahead of rear axle (68 % of weight on drive wheels)
- Band stores 31 mJ; 2.5 m needs ≈ 16 mJ $\rightarrow 1.9\times$ margin
- Predicted run ≈ 4 m, top speed ≈ 1.1 m/s
- Grip needed $\mu \geq 0.72$ (nitrile O-ring ≈ 0.9)
- Bought: 2 mm steel rod, 6x MR52, 4 O-rings, #16 band, line, 5 g steel ballast

RULES CHECK

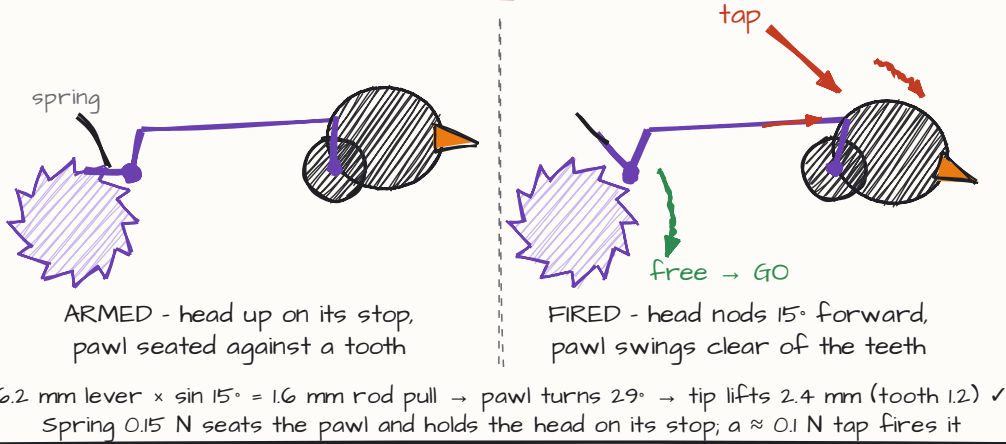
- ✓ ≥ 75 % printed PLA (78 %)
- ✓ Parts fit the egg (box C); built toy is bigger than the egg
- ✓ Stores energy and waits until the head is tapped
- ✓ Trigger is not a removable pin - nothing comes off
- ✓ No energy stored when apart (band relaxed)
- ✓ Snap / press fits - no glue, no tools needed
- ✓ No bought toy parts or downloaded CAD

A ENERGY - PULL-BACK CHARGE

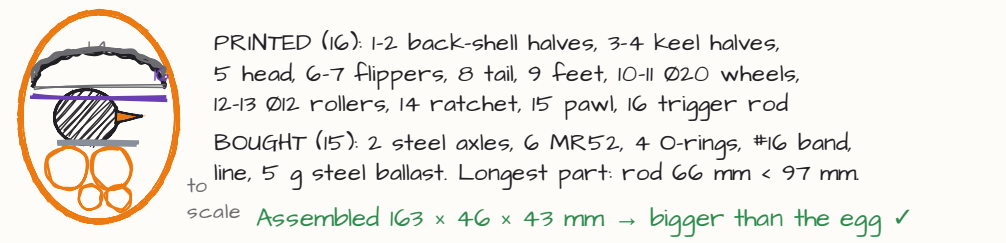


Stored $E = GA L_0 (\lambda^2/2 + 1/\lambda - 3/2) \approx 31$ mJ (G 0.45 MPa, A 2.56 mm², L_0 32 mm)
Band travel $(1.91 - 1.1) \times 32 = 26$ mm of line on the bare $\varnothing 2$ axle (r 1.2) = 22 cm pull
Peak drive 1.89 N $\times 1.2/10 = 0.23$ N vs rear load 0.32 N $\rightarrow$ needs $\mu \geq 0.72$ (O-ring ≈ 0.9)
Drag ≈ 6.5 mN (rolling 5.6, bearings + pawl 0.9) $\rightarrow 2.5$ m needs 16 mJ $\rightarrow 1.9\times$ margin
Full charge: the knot jams in the stop eyelet. Freewheel hubs let it coast after.

B TRIGGER - TAP THE HEAD



C PACKS INTO THE EGG



ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 1: sheet 1 of 1

Theme: Biomimicry - tobogganing penguin

Toy type: Running toy (≥ 2.5 m)

Scale / units: mm · side view 1:2.7 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____